

Slide 9: Five Structural Drivers

Each driver here maps directly to either a mean vector or the covariance matrix, which are the two inputs to the frontier. There are five forces explaining the shift.

On the returns side, RBI rate cuts expanded the equity risk premium. Sectoral rotations into pharma, industrials, and metals brought higher return potential compared to pre-COVID financials.

On the covariance side, sectors began moving more independently post-COVID, which reduced correlations and improved diversification. On the volatility side, FII inflows compressed idiosyncratic risk, while SIP flows absorbed market dips systematically.

With both inputs moving together, returns increased while correlations declined. This supports the interpretation of a structural shift rather than a cyclical one.

Slide 10: Biggest Winners

BHEL swung by 67 percentage points, from negative 21% to positive 47%, driven by the government's capex and PLI push.

Tata Power, Vedanta, M&M, and Sun Pharma reflect the same pattern: direct policy beneficiaries or firms aligned with global themes where India holds a strong position.

Slide 11: Pre-COVID Stars That Faded

The underperformers post-COVID are precisely the pre-COVID tangency portfolio: HDFC Bank, HUL, Asian Paints, and Kotak, each declining by 8 to 17 percentage points.

Stretched valuations, rising competition, and rate normalization eroded their premium.

This systematic reversal, where pre-COVID winners became post-COVID laggards, provides strong evidence of a regime change.

Slide 12: Limitations

Several limitations remain.

Survivorship bias is the most significant, as the fixed universe excludes failed or delisted firms, artificially improving the pre-COVID frontier.

The long-only constraint rules out short-selling, so the true attainable frontier is wider.

Transaction costs are not modeled, and the implied turnover from the tangency shift would partially reduce the Sharpe improvement.

Finally, causation cannot be established since the structural drivers are correlational.

Slide 13: Future Work

Four extensions are especially relevant for future work.

The Jobson-Korkie test would provide a p-value for the Sharpe ratio difference, allowing assessment of statistical significance.

The De Roon-Nijman test would determine whether the post-COVID frontier truly lies outside the pre-COVID frontier or reflects a reweighting of the same assets.

Frontier shift decomposition would isolate the contributions of higher returns versus improved covariance.

Bootstrap confidence bands would help assess whether the observed gap reflects a genuine effect or sampling variation.